Status	Finished
Started	Wednesday, 28 October 2026, 9:20 PM
Completed	Wednesday, 28 October 2026, 9:24 PM
Duration	3 mins 19 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting decreases the efficiency of selection operations
c. Sorting is rarely used in programming
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **2**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Merging
c. Sorting
d. Searching

The correct answer is:
Complexity

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Linear Search
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes the data unsorted
c. It complicates the selection of items
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **5**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Arranging data in a particular format
c. Inserting data into a list
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **6**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Dividing the list into halves
c. Sorting the entire list in one pass
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **7**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Insertion
b. Selection
c. Distribution
d. Extraction

The correct answer is:
Insertion

Question **8**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sort()
c. sorted()
d. order()

The correct answer is: sorted()

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method

b. Divide and conquer

c. Backtracking

d. Dynamic programming

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort

b. It is based on the divide-and-conquer approach

c. It does not require recursion

d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It bubbles up the largest element in each pass

b. It uses the heap data structure

c. It divides the list into sublists

d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **12**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It has no effect on the analysis process
c. It complicates the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Bubble Sort
c. Insertion Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Merge sort
b. Selection sort
c. Heap sort
d. Quick sort

The correct answer is:
Merge sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Quick Sort
d. Insertion Sort

The correct answer is: Bubble Sort